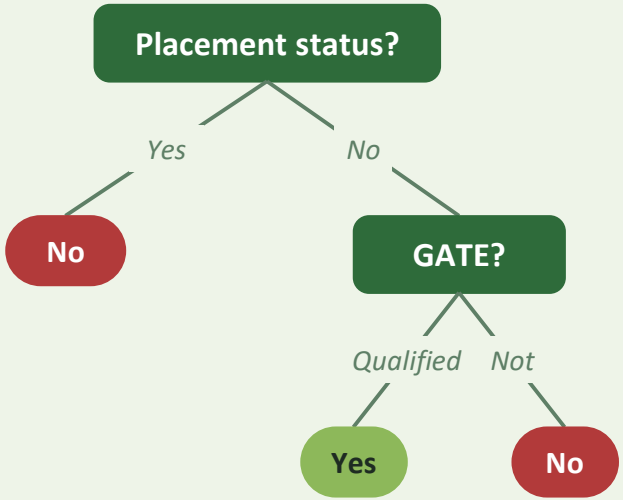


Decision Tree Classification

What is a decision tree?

- A supervised learning model that predicts an outcome by asking a sequence of questions about the features.
- Used for both classification (Yes/No, class labels) and regression (numeric values).
- Each answer moves you down one branch until you reach a final decision.
- Easy to read and explain — the model is literally a flowchart of if/else rules.

Example: Should I continue study in a college?



Anatomy of a decision tree

1

Root node

The first split — the attribute with the highest information gain.

2

Decision node

An internal test on an attribute (drawn as a box).

3

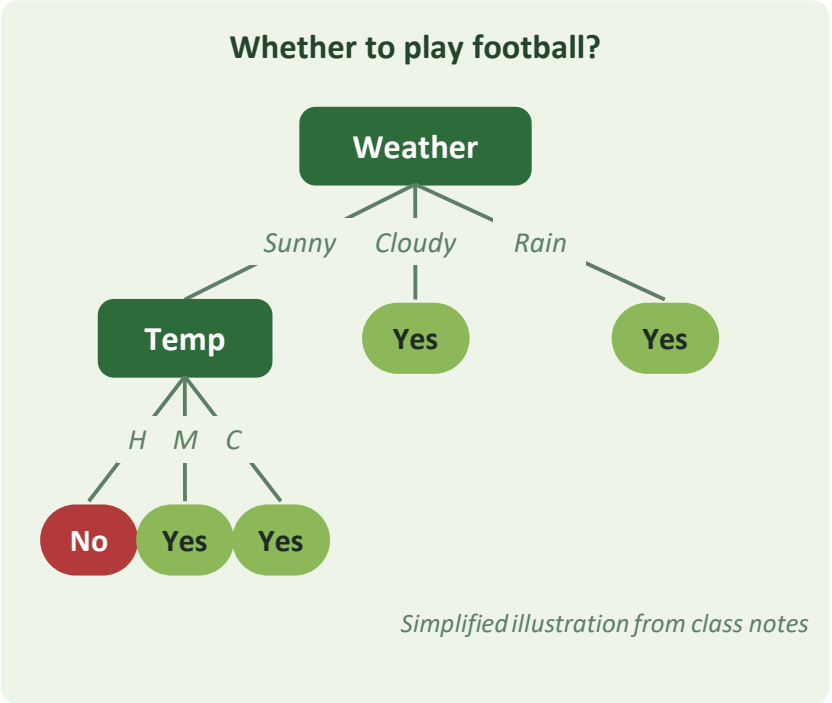
Branch

One possible value of the attribute being tested.

4

Leaf node

The final outcome; ideally a pure class (all Yes or all No).



Simplified illustration from class notes

Measuring purity: Entropy and Info Gain

Entropy

$$E(S) = -\,p_{+}\log_2 p_{+} - p_{-}\log_2 p_{-}$$

How mixed a set is. 0 = pure (all one class); 1 = a 50/50 split between two classes.

Information Gain

$$IG(S, A) = E(S) - \Sigma\,(|S_v| \,/\, |S|) \cdot E(S_v)$$

The drop in entropy after splitting on attribute A. Higher IG = purer children = better split.

1. Compute entropy of the dataset
2. Compute IG for every attribute
3. Split on the highest-IG attribute
4. Repeat on each branch until leaves are pure

The ID3 algorithm

The dataset: Should we play football?

Day	Weather	Temp	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Cloudy	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Cloudy	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Cloudy	Mild	High	Strong	Yes
13	Cloudy	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

14 days · 9 Yes · 5 No

Steps to build the tree

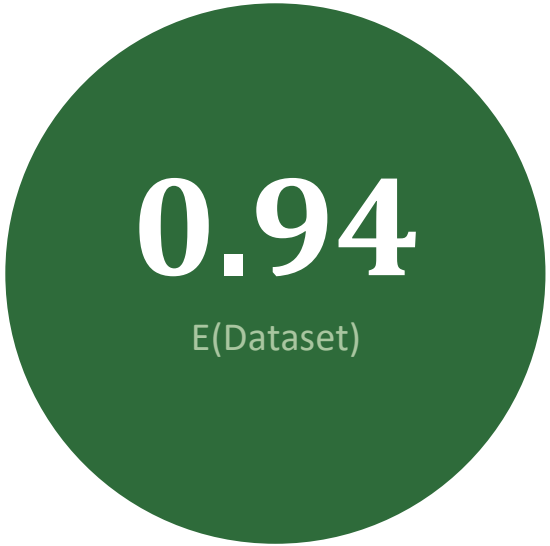
1. Find the entropy of the dataset
2. Find the IG of every attribute
3. Highest IG becomes the root node
4. Make a new dataset for each branch
5. Repeat IG on each new dataset
6. Stop when every leaf is pure

Step 1: Entropy of the whole dataset

S = {+9, −5} (9 Yes, 5 No out of 14)

$$E(S) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14)$$
$$= 0.410 + 0.530$$
$$= \mathbf{0.940}$$

The dataset is quite mixed (close to 1), so there is plenty of uncertainty for a good split to remove.



Step 2: Information Gain of Weather

Sunny

5 days · {+2, −3}

$-(2/5)\log_2(2/5) - (3/5)\log_2(3/5)$

E = 0.971

Cloudy

4 days · {+4, −0}

$-(4/4)\log_2(4/4)$

E = 0 (pure)

Rain

5 days · {+3, −2}

$-(3/5)\log_2(3/5) - (2/5)\log_2(2/5)$

E = 0.971

$$IG(S, Weather) = E(S) - (5/14)\cdot E(Sunny) - (4/14)\cdot E(Cloudy) - (5/14)\cdot E(Rain)$$

$$= 0.940 - (5/14)(0.971) - (4/14)(0) - (5/14)(0.971) = \mathbf{0.246}$$

Step 3: Information Gain of Temperature

Hot

4 days · {+2, −2}

$-(2/4)\log_2(2/4) - (2/4)\log_2(2/4)$

E = 1.000

Mild

6 days · {+4, −2}

$-(4/6)\log_2(4/6) - (2/6)\log_2(2/6)$

E = 0.918

Cool

4 days · {+3, −1}

$-(3/4)\log_2(3/4) - (1/4)\log_2(1/4)$

E = 0.811

$$IG(S, Temp) = E(S) - (4/14)\cdot E(Hot) - (6/14)\cdot E(Mild) - (4/14)\cdot E(Cool)$$

$$= 0.940 - (4/14)(1.000) - (6/14)(0.918) - (4/14)(0.811) = \mathbf{0.029}$$

Step 4: Information Gain of Humidity

High

7 days · {+3, −4}

$-(3/7)\log_2(3/7) - (4/7)\log_2(4/7)$

E = 0.985

Normal

7 days · {+6, −1}

$-(6/7)\log_2(6/7) - (1/7)\log_2(1/7)$

E = 0.592

$$IG(S, Humidity) = E(S) - (7/14)\cdot E(High) - (7/14)\cdot E(Normal)$$

$$= 0.940 - (7/14)(0.985) - (7/14)(0.592) = \mathbf{0.152}$$

Step 5: Information Gain of Wind

Weak

8 days · {+6, −2}

$-(6/8)\log_2(6/8) - (2/8)\log_2(2/8)$

E = 0.811

Strong

6 days · {+3, −3}

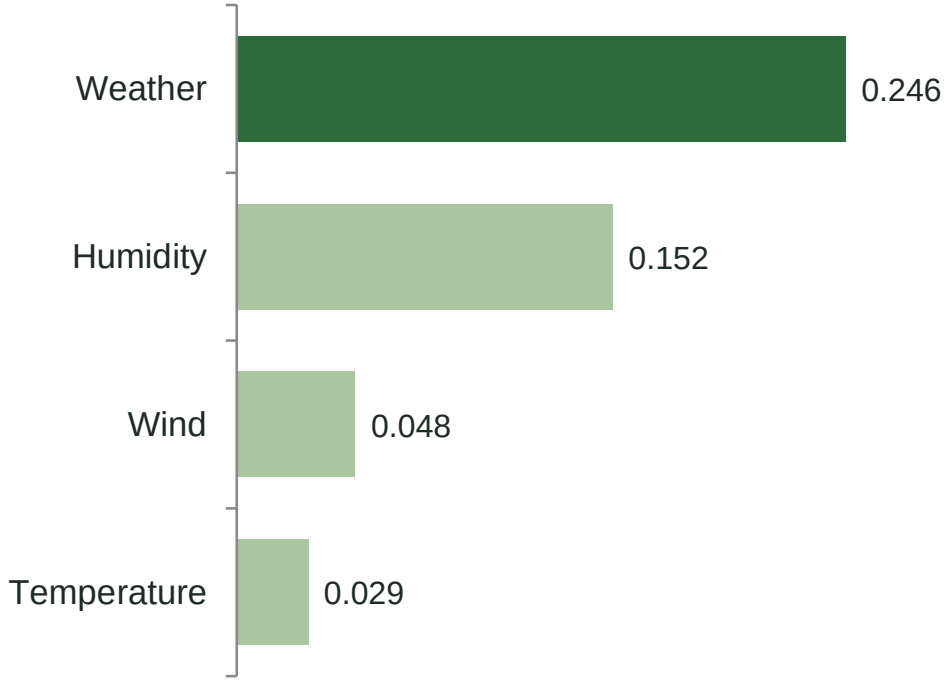
$-(3/6)\log_2(3/6) - (3/6)\log_2(3/6)$

E = 1.000

$IG(S, Wind) = E(S) - (8/14)\cdot E(Weak) - (6/14)\cdot E(Strong)$

$= 0.940 - (8/14)(0.811) - (6/14)(1.000) =$ **0.048**

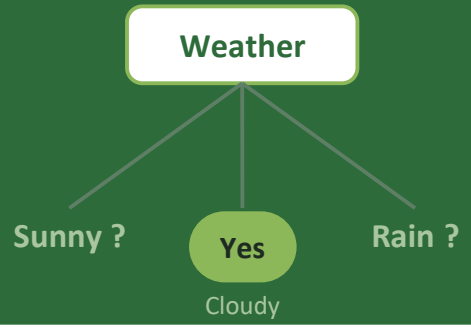
Step 6: Compare gains and select the root



Max IG = 0.246

Weather is the root node

It removes the most uncertainty, and its Cloudy branch is already pure (all Yes).



Next node: the new dataset for Sunny

Day	Weather	Temp	Humidity	Wind	Play?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes

Only the 5 Sunny days are kept. Weather is already used, so the candidates are Temperature, Humidity and Wind.

5 Sunny days

2 Yes · 3 No

$$E(\text{Sunny}) = -(2/5)\log_2(2/5) - (3/5)\log_2(3/5)$$

= 0.971

Sunny: Information Gain of Temperature

Hot

Days 1, 2 · {+0, −2}

$-(2/2)\log_2(2/2)$

E = 0 (pure)

Mild

Days 8, 11 · {+1, −1}

$-(1/2)\log_2(1/2) - (1/2)\log_2(1/2)$

E = 1.000

Cool

Day 9 · {+1, −0}

$-(1/1)\log_2(1/1)$

E = 0 (pure)

$$IG(Sunny, Temp) = E(Sunny) - (2/5)\cdot E(Hot) - (2/5)\cdot E(Mild) - (1/5)\cdot E(Cool)$$

$$= 0.971 - (2/5)(0) - (2/5)(1.000) - (1/5)(0) = \mathbf{0.571}$$

Sunny: Information Gain of Humidity

High

Days 1, 2, 8 · {+0, −3}

$-(3/3)\log_2(3/3)$

E = 0 (pure)

Normal

Days 9, 11 · {+2, −0}

$-(2/2)\log_2(2/2)$

E = 0 (pure)

$$IG(Sunny, Humidity) = E(Sunny) - (3/5)\cdot E(High) - (2/5)\cdot E(Normal)$$

$$= 0.971 - (3/5)(0) - (2/5)(0) = \mathbf{0.971}$$

Sunny: Information Gain of Wind

Weak

Days 1, 8, 9 · {+1, −2}

$-(1/3)\log_2(1/3) - (2/3)\log_2(2/3)$

E = 0.918

Strong

Days 2, 11 · {+1, −1}

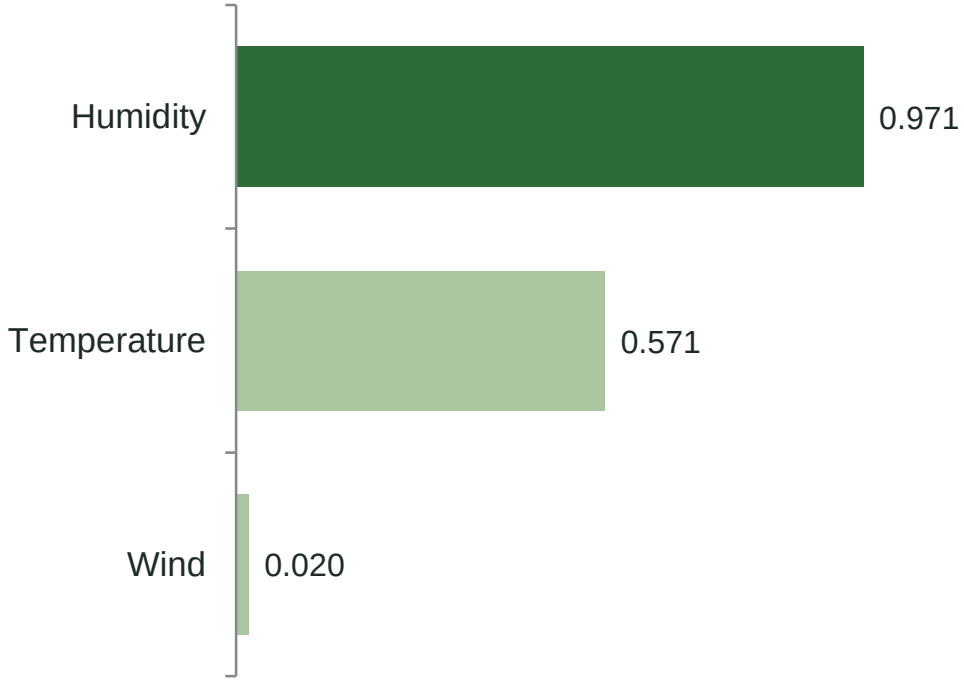
$-(1/2)\log_2(1/2) - (1/2)\log_2(1/2)$

E = 1.000

$$IG(Sunny, Wind) = E(Sunny) - (3/5)\cdot E(Weak) - (2/5)\cdot E(Strong)$$

$$= 0.971 - (3/5)(0.918) - (2/5)(1.000) = \mathbf{0.020}$$

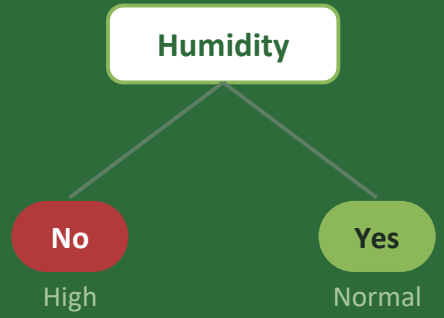
Sunny: compare gains and split on Humidity



Max IG = 0.971

Split Sunny on Humidity

Both branches are pure, so they become leaf nodes straight away.



Next node: the new dataset for Rain

Day	Weather	Temp	Humidity	Wind	Play?
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
10	Rain	Mild	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

Only the 5 Rain days are kept. Weather is already used, so the candidates are Temperature, Humidity and Wind. (Cloudy needs no split: it is already pure, all Yes.)

5 Rain days

3 Yes · 2 No

$$E(\text{Rain}) = -(3/5)\log_2(3/5) - (2/5)\log_2(2/5)$$

= 0.971

Rain: Information Gain of Temperature

Hot

No Rain days · {+0, −0}

Empty subset, weight 0/5

E = 0 (no data)

Mild

Days 4, 10, 14 · {+2, −1}

$-(2/3)\log_2(2/3) - (1/3)\log_2(1/3)$

E = 0.918

Cool

Days 5, 6 · {+1, −1}

$-(1/2)\log_2(1/2) - (1/2)\log_2(1/2)$

E = 1.000

$$IG(Rain, Temp) = E(Rain) - (0/5)\cdot E(Hot) - (3/5)\cdot E(Mild) - (2/5)\cdot E(Cool)$$

$$= 0.971 - 0 - (3/5)(0.918) - (2/5)(1.000) = \mathbf{0.020}$$

Rain: Information Gain of Humidity

High

Days 4, 14 · {+1, −1}

$-(1/2)\log_2(1/2) - (1/2)\log_2(1/2)$

E = 1.000

Normal

Days 5, 6, 10 · {+2, −1}

$-(2/3)\log_2(2/3) - (1/3)\log_2(1/3)$

E = 0.918

$$IG(Rain, Humidity) = E(Rain) - (2/5)\cdot E(High) - (3/5)\cdot E(Normal)$$

$$= 0.971 - (2/5)(1.000) - (3/5)(0.918) = \mathbf{0.020}$$

Rain: Information Gain of Wind

Weak

Days 4, 5, 10 · {+3, −0}

$-(3/3)\log_2(3/3)$

E = 0 (pure)

Strong

Days 6, 14 · {+0, −2}

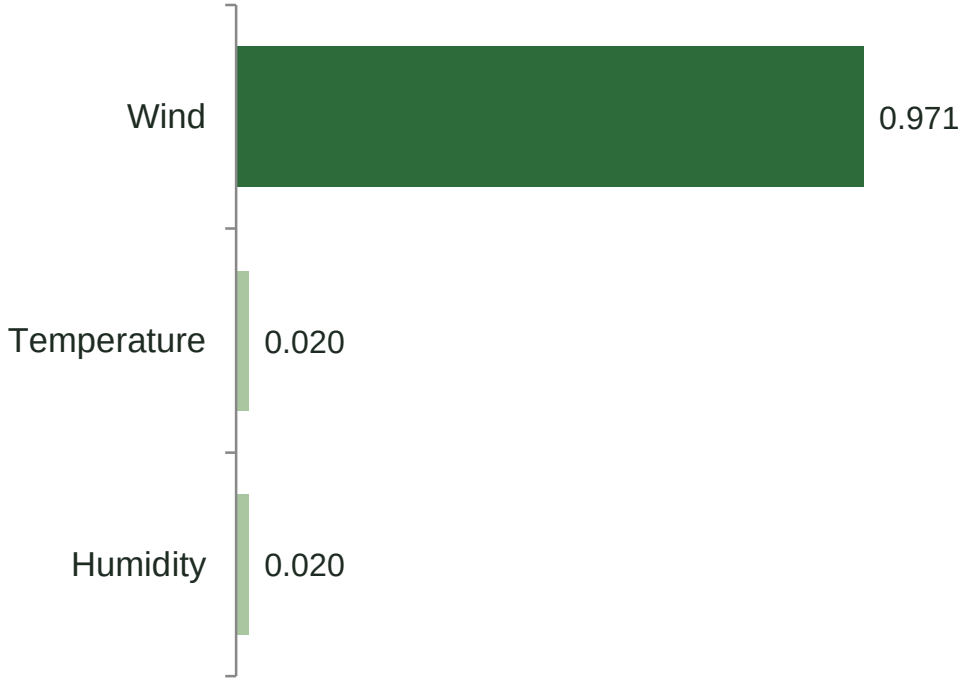
$-(2/2)\log_2(2/2)$

E = 0 (pure)

$$IG(Rain, Wind) = E(Rain) - (3/5)\cdot E(Weak) - (2/5)\cdot E(Strong)$$

$$= 0.971 - (3/5)(0) - (2/5)(0) = \mathbf{0.971}$$

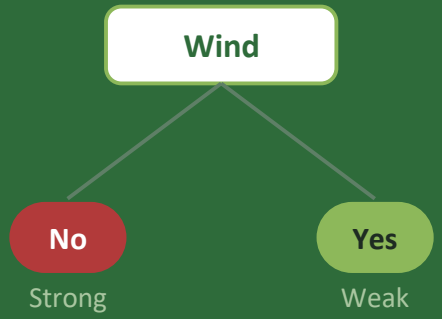
Rain: compare gains and split on Wind



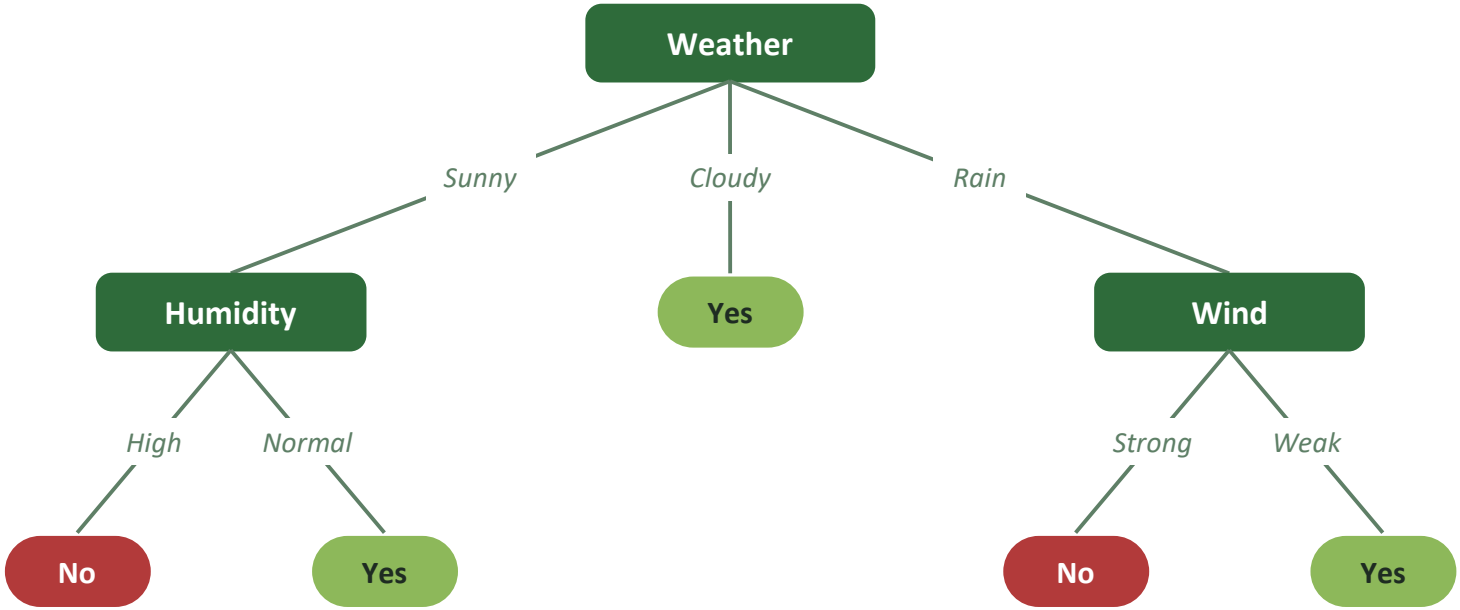
Max IG = 0.971

Split Rain on Wind

Both branches are pure, so they become leaf nodes straight away.



The final decision tree



Every leaf is pure, so the tree classifies all 14 training days correctly. Temperature was never needed.

Exercise: decision tree for A = (X AND Y) OR Z

X	Y	Z	A
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

5 True, 3 False

$E(S) = -(5/8)\log_2(5/8) - (3/8)\log_2(3/8) = 0.954$

1. Root: IG on all 8 rows

Z=1 {4T, 0F}: E=0 · Z=0 {1T, 3F}: E=0.811

IG(Z) = 0.954 – 0 – (4/8)(0.811) = 0.549

X=1 {3T,1F}: 0.811 · X=0 {2T,2F}: 1.0

IG(X) = IG(Y) = 0.954 – 0.406 – 0.5 = 0.049

Highest IG → Z is the root

2. Branch Z = 0 (4 rows: 1T, 3F)

$E(Z=0) = -\frac{1}{4}\log_2\frac{1}{4} - \frac{3}{4}\log_2\frac{3}{4} = 0.811$

X=1 {1T,1F}: 1.0 · X=0 {0T,2F}: 0

IG(X) = IG(Y) = 0.811 – (2/4)(1.0) = 0.311

Tie → choose X; X=0 is pure (False)

X=1 → split on Y: Y=1 True, Y=0 False

